



Advanced Digital Image Processing Lecture #1

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Purpose of This Course

- ❖ Improve insight for image processing (IP)
- ❖ Find proper ways to apply IP techniques to your particular problem
- ❖ learn how to read IP papers in proper viewpoints
- ❖ Points: representation, model, and process
- ❖ Reason: human sensation and physical imaging system



Configuration of the Course

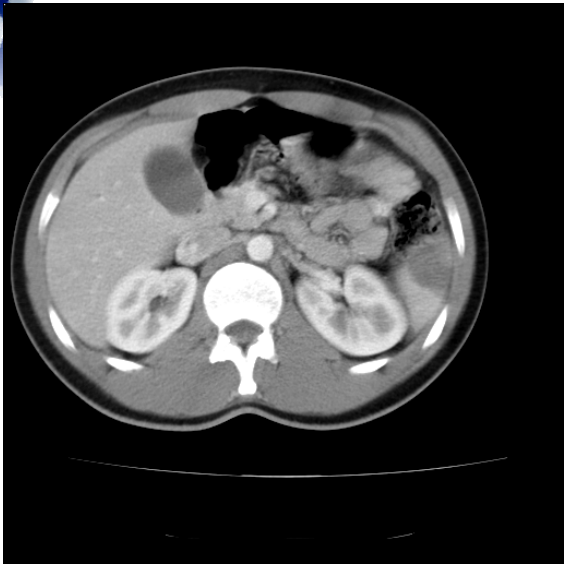
- ❖ 15 classes are planned
- ❖ Time for each class: 100 min
- ❖ Student presentation: on good IP papers or specific topics
- ❖ Credits: awarded by presentation, midterm examination and final examination



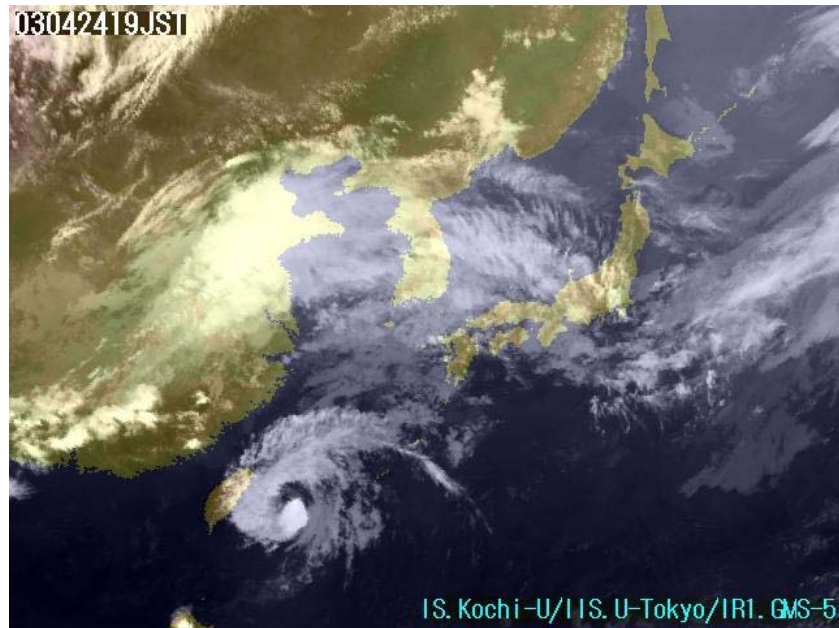
Purpose of IP

- ❖ Visualize invisible information
- ❖ Image conversion (enhancement, restoration, etc.)
- ❖ Image Analysis and object recognition
- ❖ Delineate symbolic information from images (OCR, robot vision, etc.)

Visualize Invisible Information

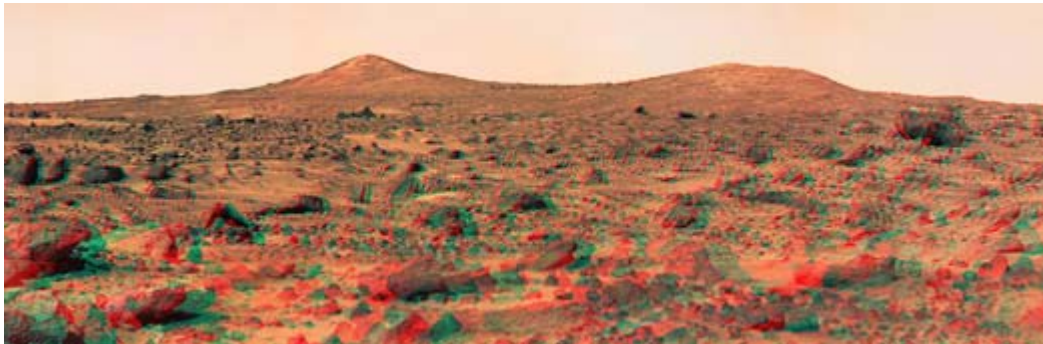


CT image



Infrared image

Image Restoration and Enhancement



Mars Pathfinder image (Super resolution)

Symbolic Information Extraction

82771 13211 092 73167
82771 (92771) 13211 (13210) 092 (892) 73167 (73169)

12701 6963 698411 5821
12701 (12701) 6963 (8983) 698411 (690411) 5821 (5827)

407599 23776 86 07 550
407599 (40059) 23776 (23976) 86 (56) 07 (01) 550 (80)

102299 108 99293 5720
102299 (62299) 108 (68) 99293 (79293) 5720 (5420)

(a)

226440 7883 45537 5649
226440 7883 45537 5649

71329 43733 87251 3976
71329 43733 87251 3976

4507 29344 83830 552
4507 29344 83830 552

9048 2522 06433 96 5434
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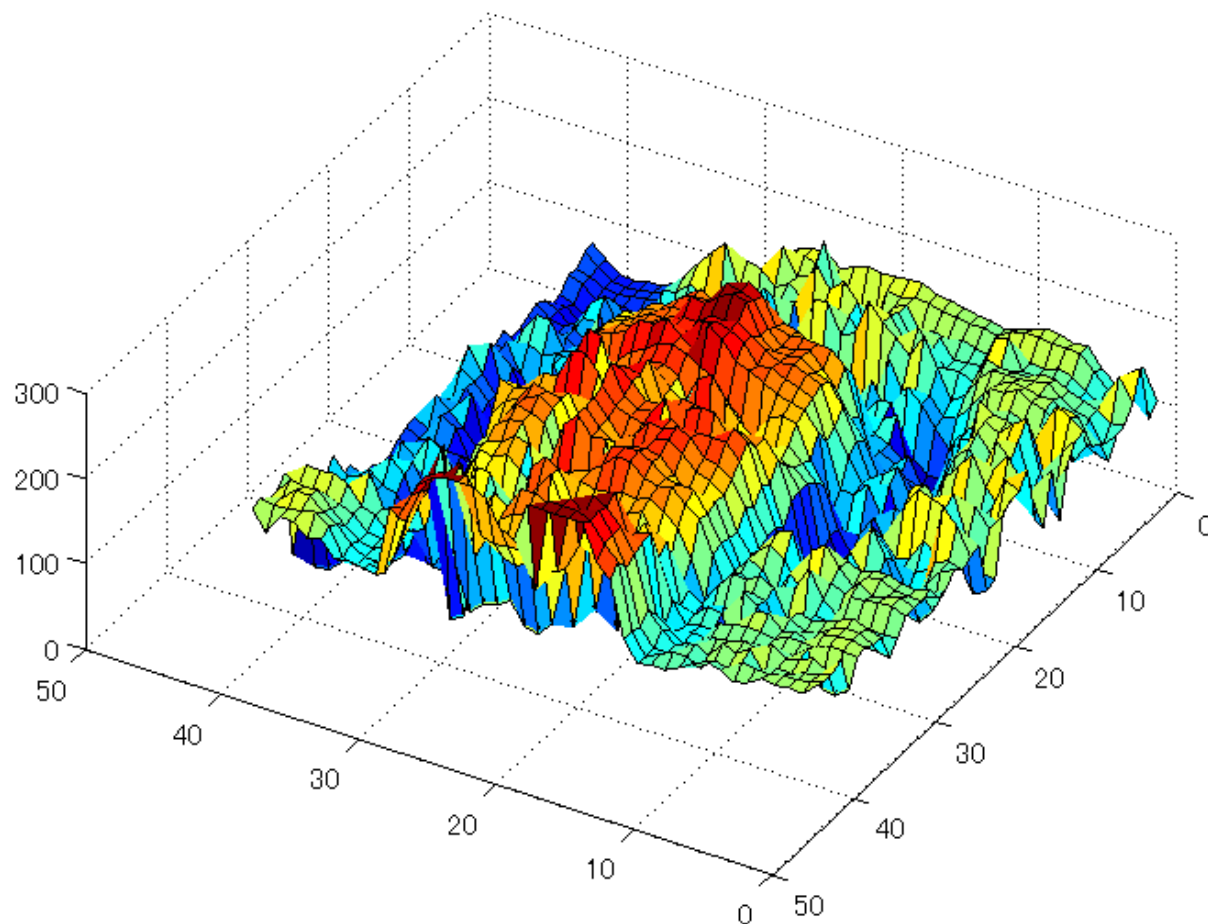
(b)



Why Image?

- ❖ Human has too good ability in image processing.
- ❖ We can get rich information from the real world from image.
- ❖ Constitution of image is mostly influenced by two aspects:
human's visual sensation system and
physical system of image formulation.

Image as Intensity Data



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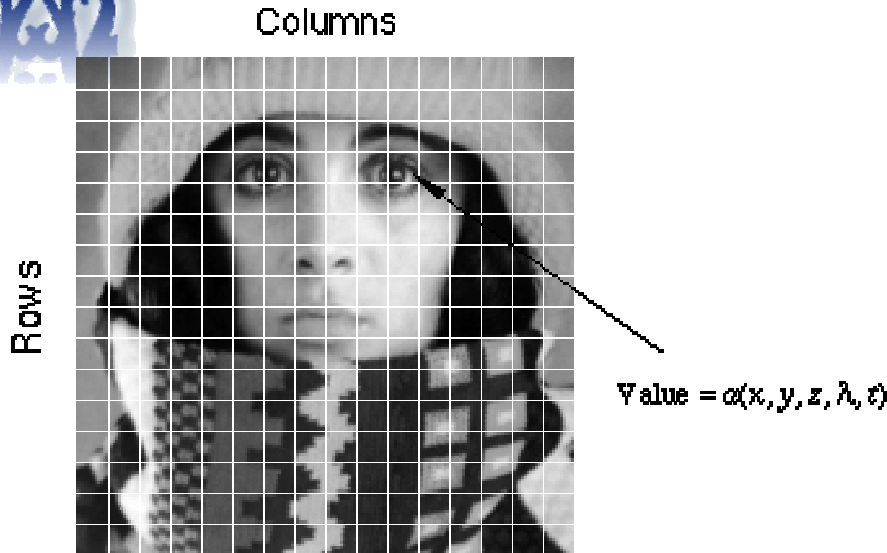
Human Visual Sensation System

- ❖ Biological constitution of eyes
- ❖ Similarity to camera system
- ❖ Color sensation (discrepancies between biological sensation and physical formulation)

Definition of Image

- ❖ An image defined in the "real world" is considered to be a function of two real variables, for example, $a(x,y)$ with a as the amplitude (e.g. brightness) of the image at the *real* coordinate position (x,y) .
- ❖ A digital image $a[m,n]$ described in a 2D discrete space is derived from an analog image $a(x,y)$ in a 2D continuous space through a *sampling* process that is frequently referred to as digitization.

Example of Digitization



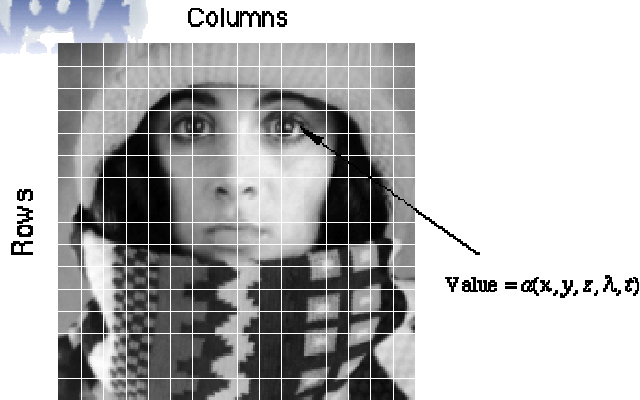
✓ The 2D continuous image $I(x,y)$ is divided into N rows and M columns.

✓ The intersection of a row and a column is termed a **pixel**.

✓ The value assigned to the integer coordinates $[m,n]$ with $\{m=0,1,2,...,M-1\}$ and $\{n=0,1,2,...,N-1\}$ is $I[m,n]$.

depth (z), color (λ), time (t)

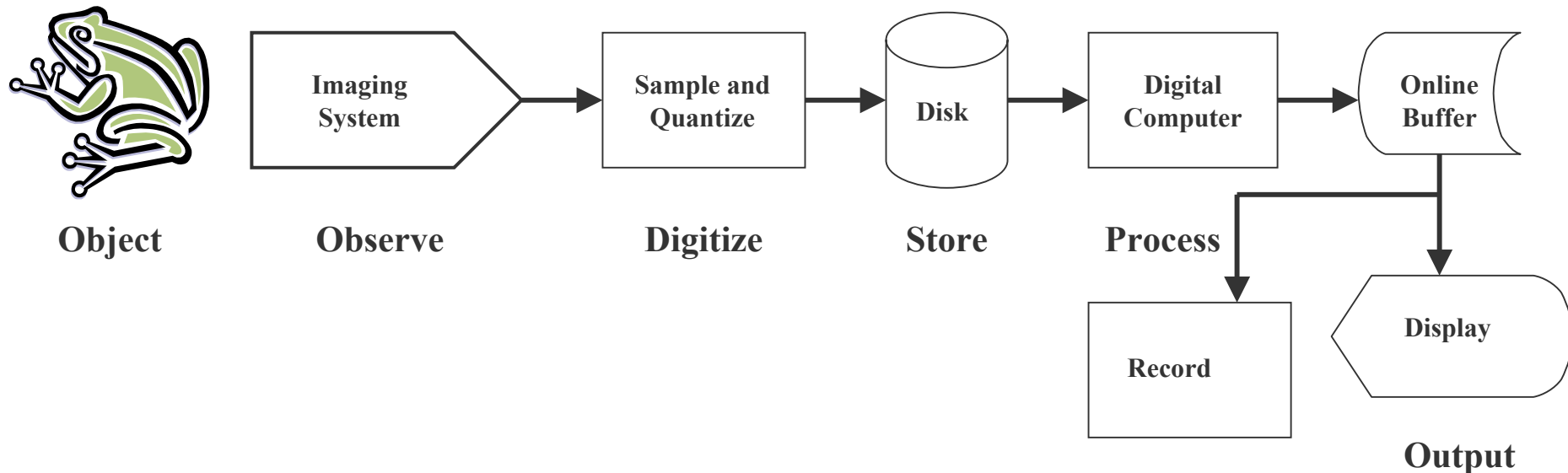
Quantization



- ❖ Image divided into $N = 16$ rows and $M = 16$ columns.

- ❖ The process of representing the amplitude of the 2D signal at a given coordinate as an integer value with L different gray levels is usually referred to as amplitude quantization or simply *quantization*
- ❖ The value assigned to every pixel is the average brightness in the pixel rounded to the nearest integer value.

Block Diagram of a Typical Image Processing Sequence





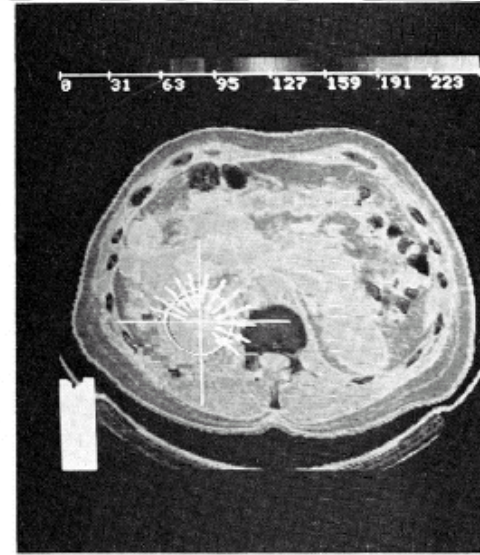
Applications of Digital Image Processing

- ❖ Remote sensing Via Satellites and other Spacecrafts
- ❖ Image Transmission and Storage for Business Applications
- ❖ Medical Image Processing
- ❖ Radar
- ❖ Robotics
- ❖ Automatic Inspection of Industrial Parts
- ❖ Person Identification for Security Measure, etc.

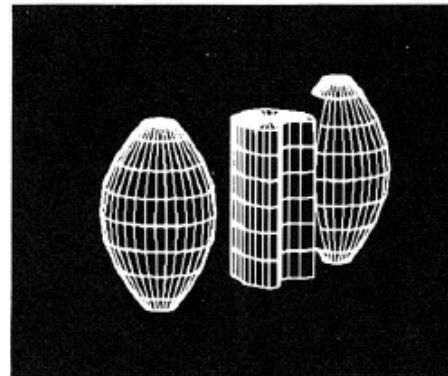
Application: Medical Image



Scan Data^(a)

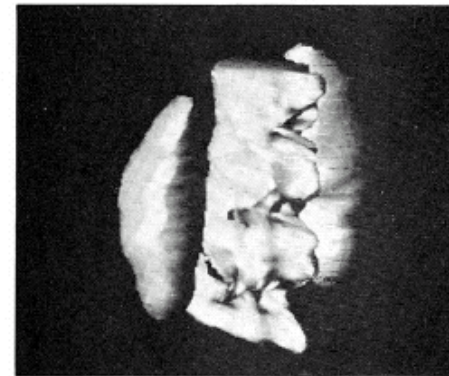


Model Fitting



(b)

Prototype Kidney Model



(d)

Resulting Kidney
and spinal cord in
stance



Application: Face Detection



Application: Human-Robot Interaction



List of Reference Books

- ❖ Rafael C. Gonzalez, “Digital Image Processing”.
- ❖ Anil K. Jain, “Fundamental of Digital Image Processing”.
- ❖ Dana H. Ballard, “Computer Vision”
- ❖ David H. Hubel, “Eye, Brain, and Vision”
- ❖ Dwayne Phillips, “Image Processing in C”

Recommended Journals to Study

- ❖ IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)
- ❖ IEEE Trans. on Image Processing (ToIP)
- ❖ International Journal on Computer Vision (IJCV)
- ❖ Computer Vision and Image Understanding (CVIU)
- ❖ IEEE Trans. on Multimedia (TMM)



End of Presentation

Thanks to all !!!